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(54) **DISCRETE ROLLER ASSEMBLY-BASED
HELICAL DRIVE FOR TRANSLATION**

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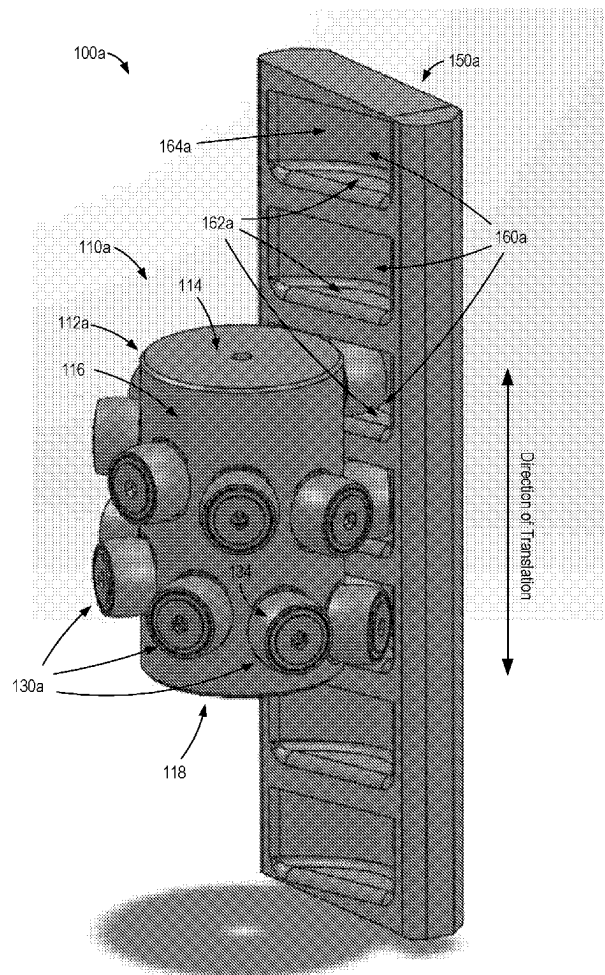
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(57)

ABSTRACT

Systems and methods for operation and assembly of a helical drive are provided. A helical drive includes a discrete roller assembly-based roller worm including a body portion and a plurality of discrete roller assemblies disposed in a first helical pattern about the body portion. The helical drive includes an engagement structure including a primary axis, wherein the roller worm is configured to rotate about a central axis to cause linear translation of the roller worm along a length of the engagement structure and the primary axis. The engagement structure includes a plurality of tracks disposed along the length of the engagement structure, where each of the tracks is configured to receive at least one of the roller assemblies during the rotation of the roller worm.



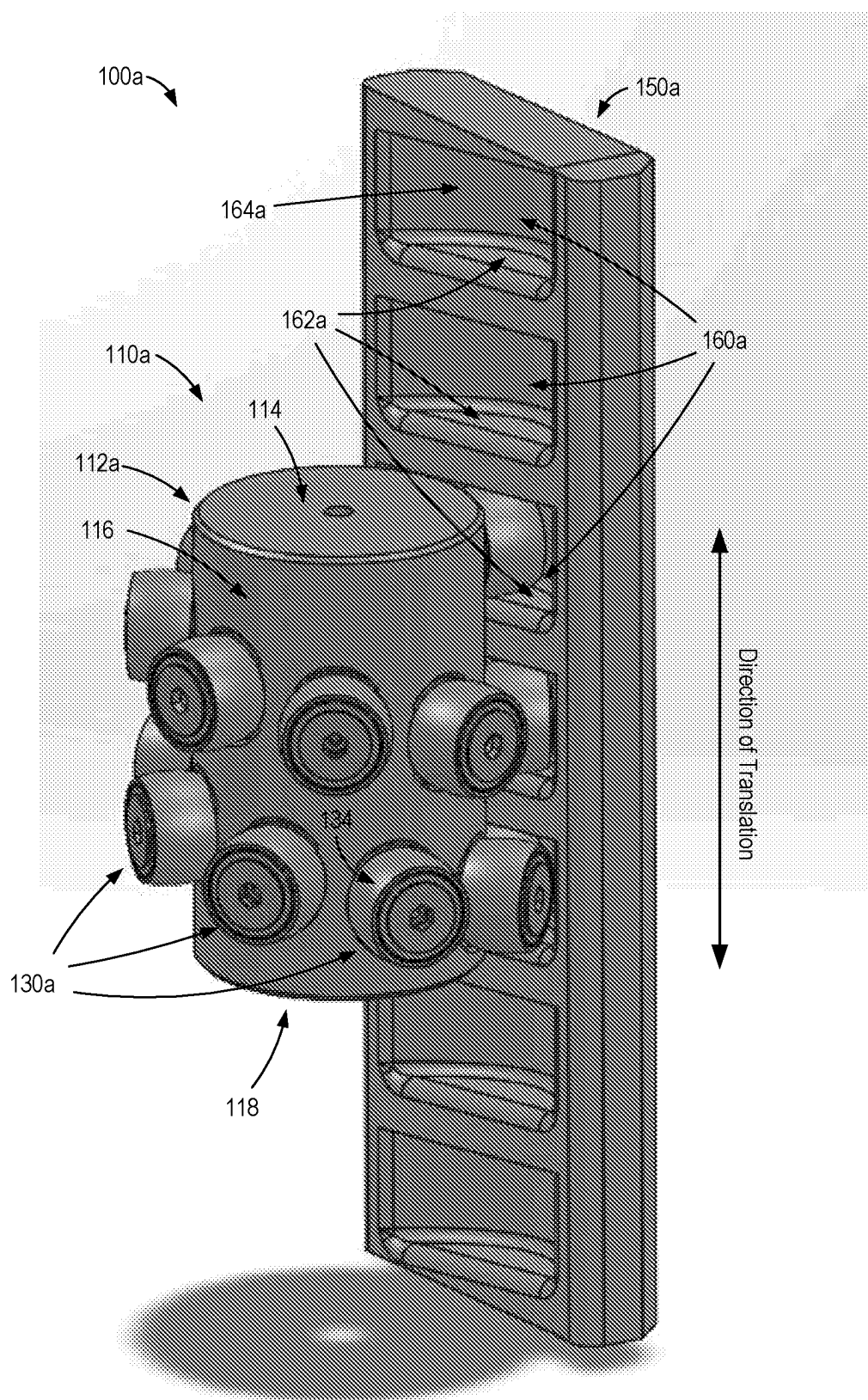


FIG. 1A

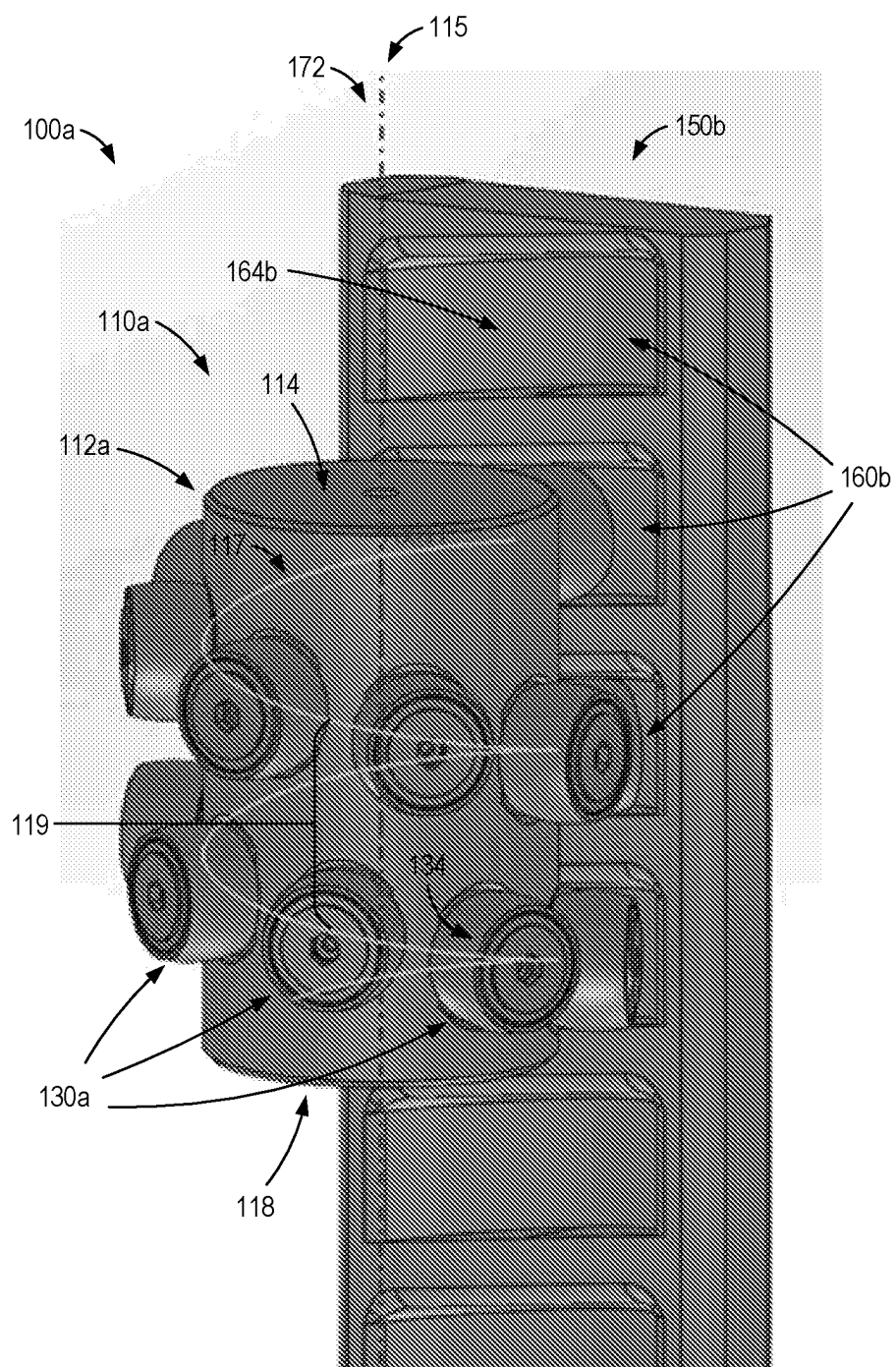


FIG. 1B

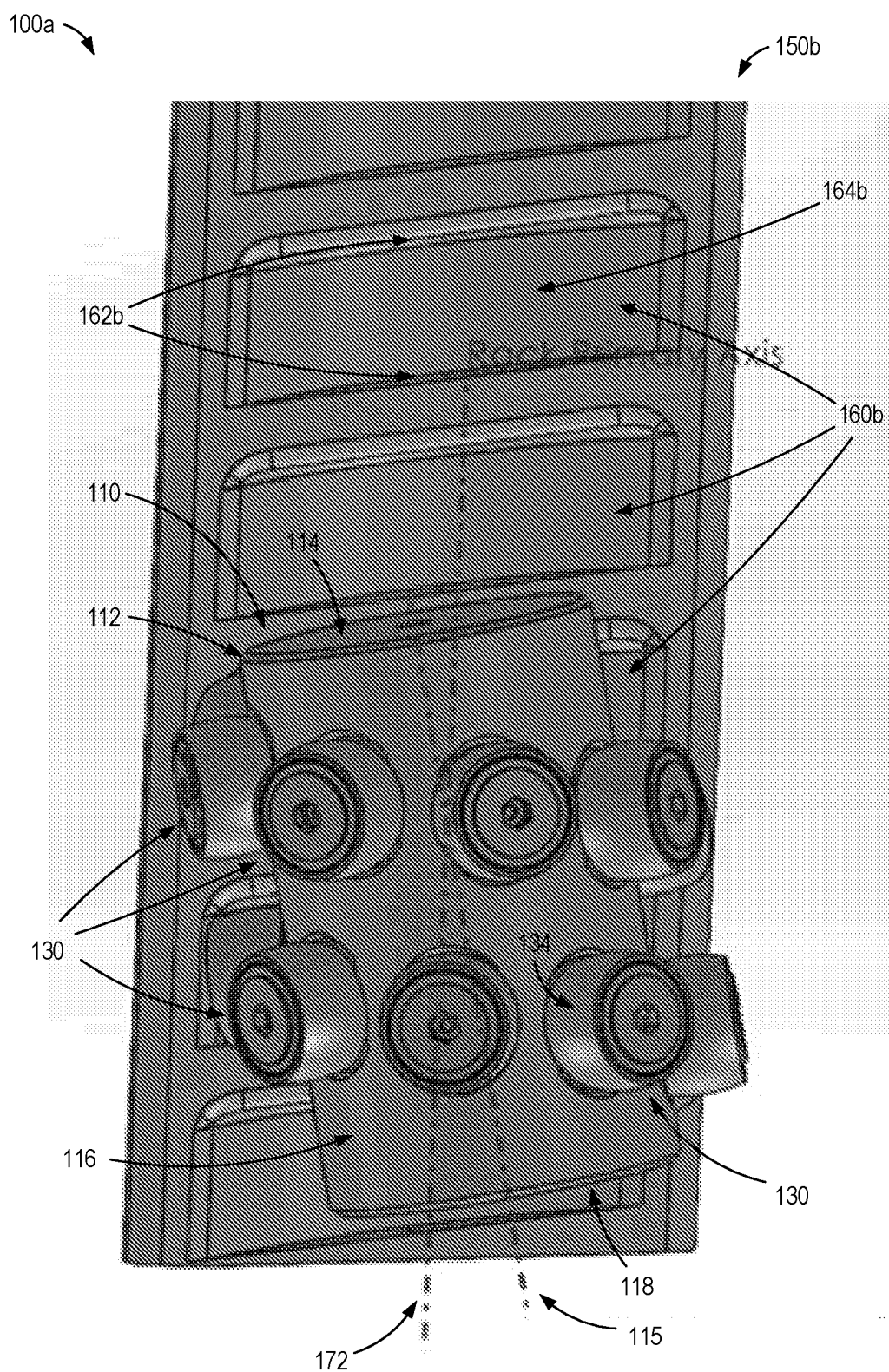


FIG. 1C

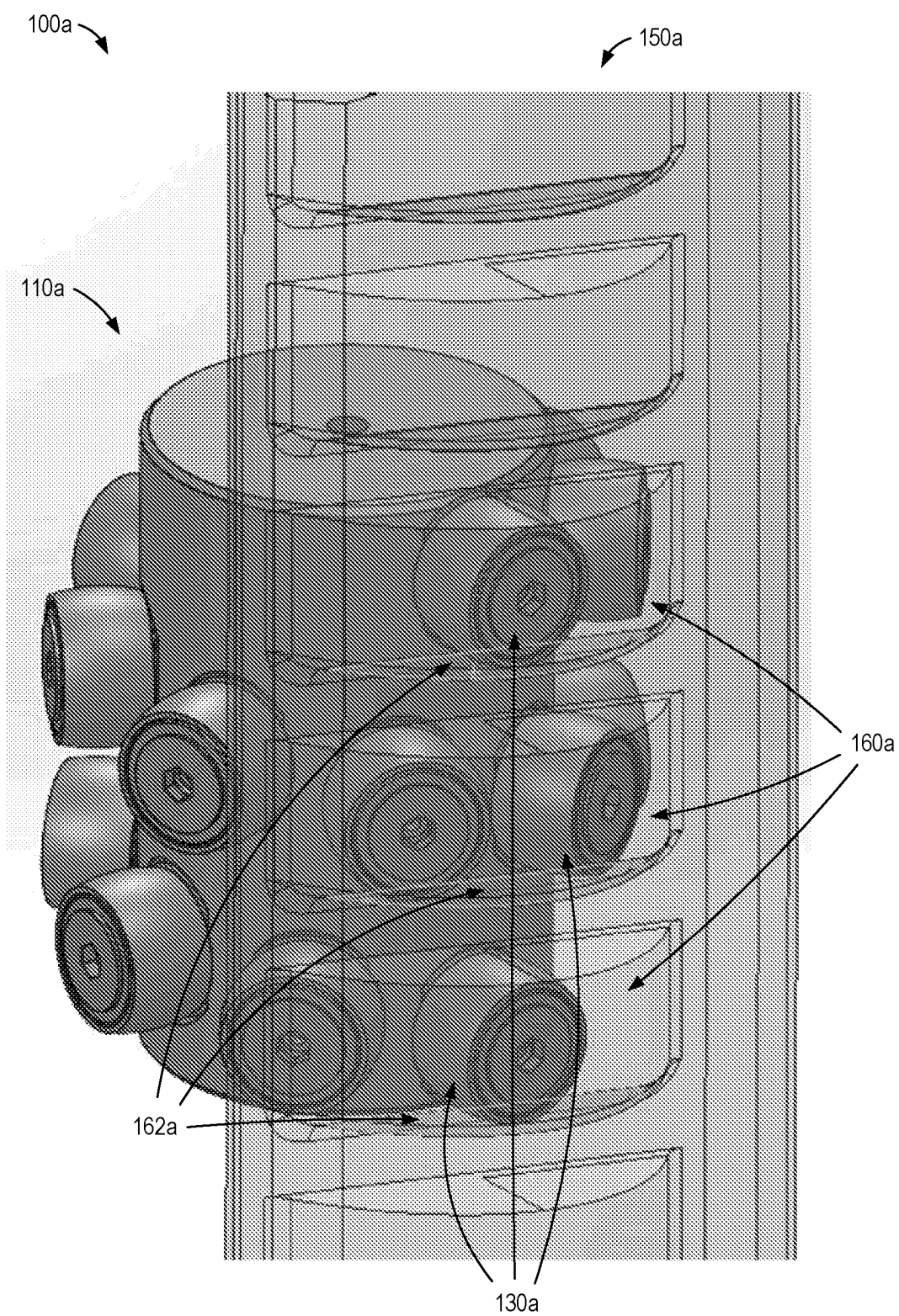


FIG. 1D

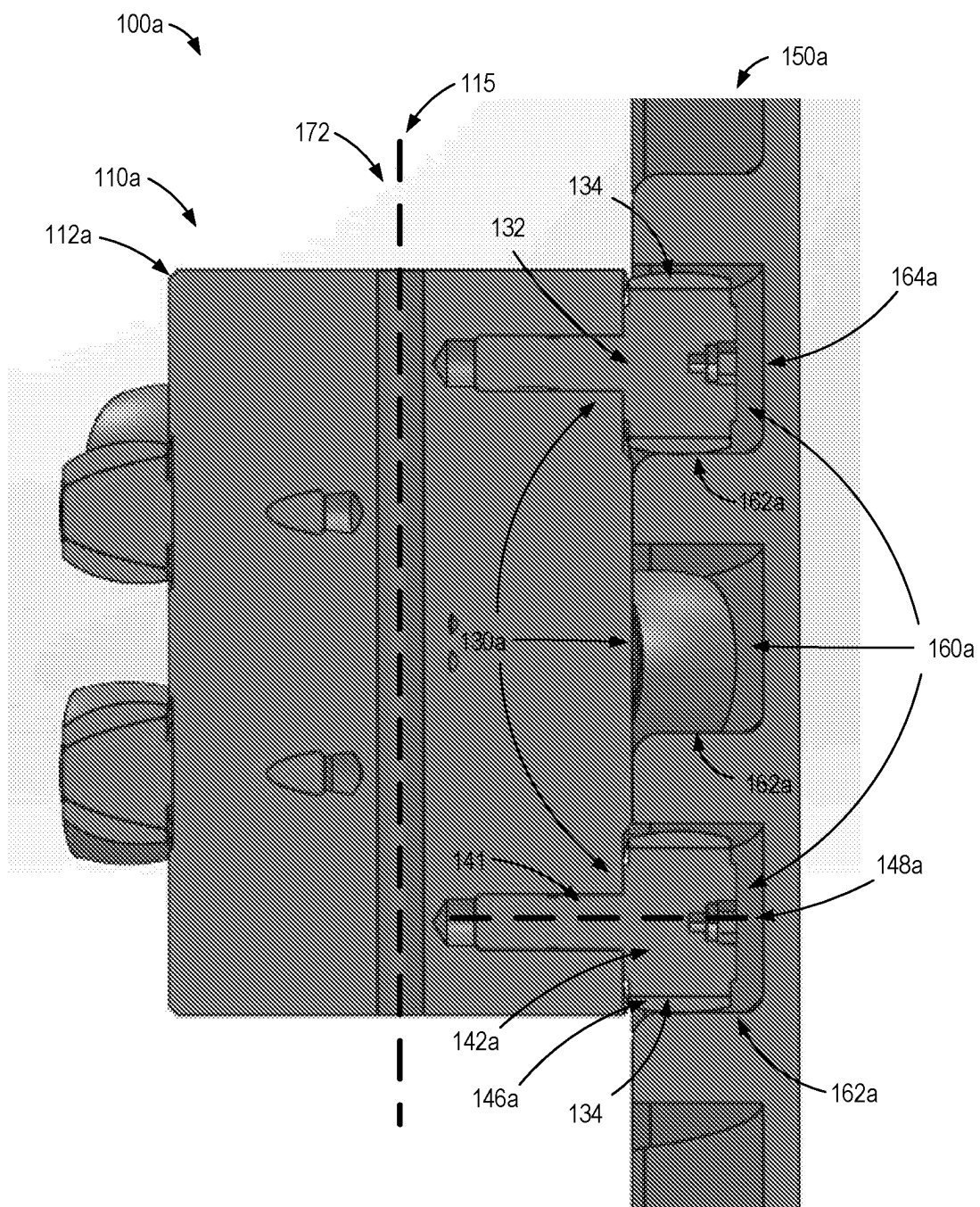


FIG. 1E

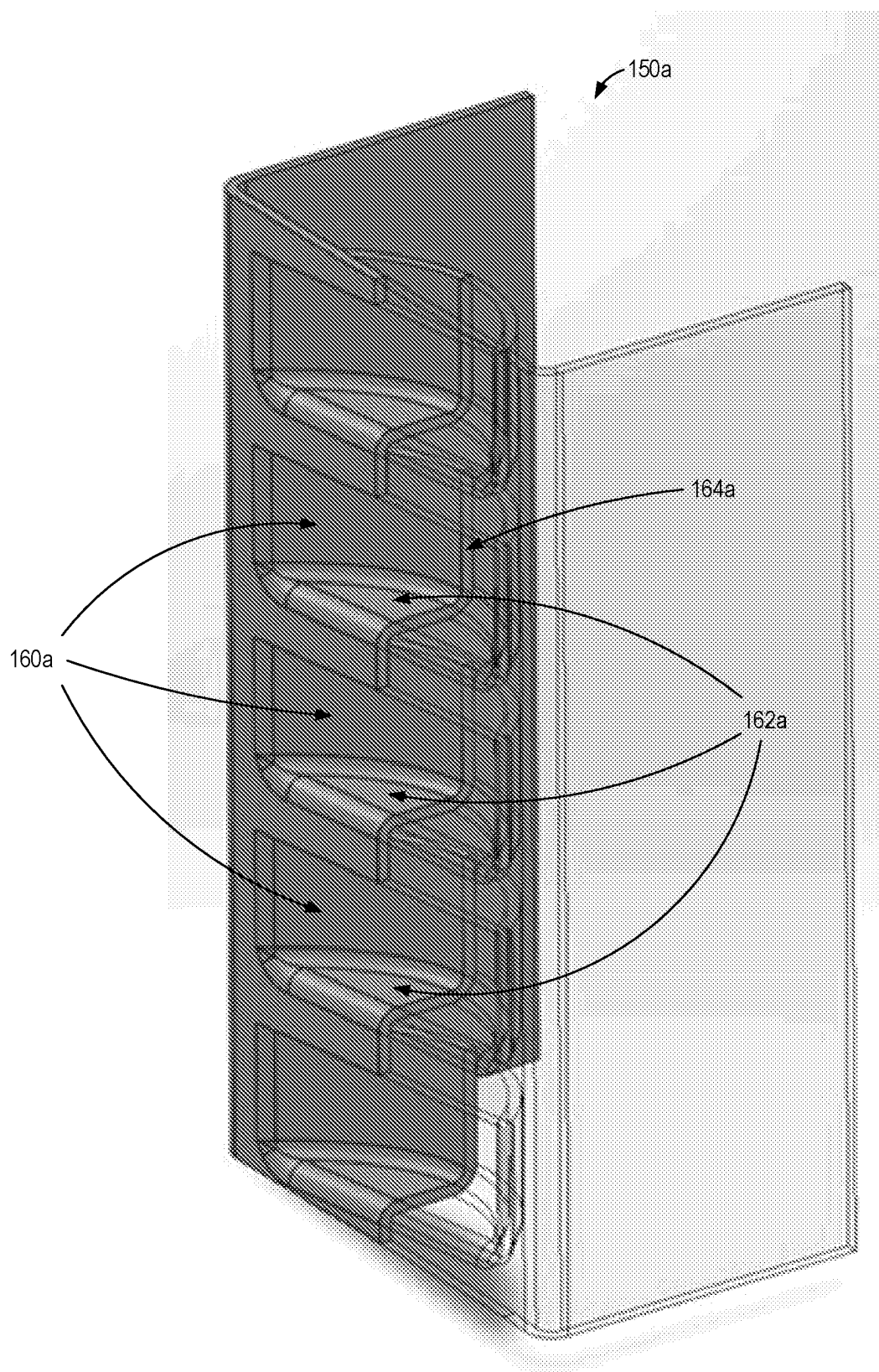


FIG. 2

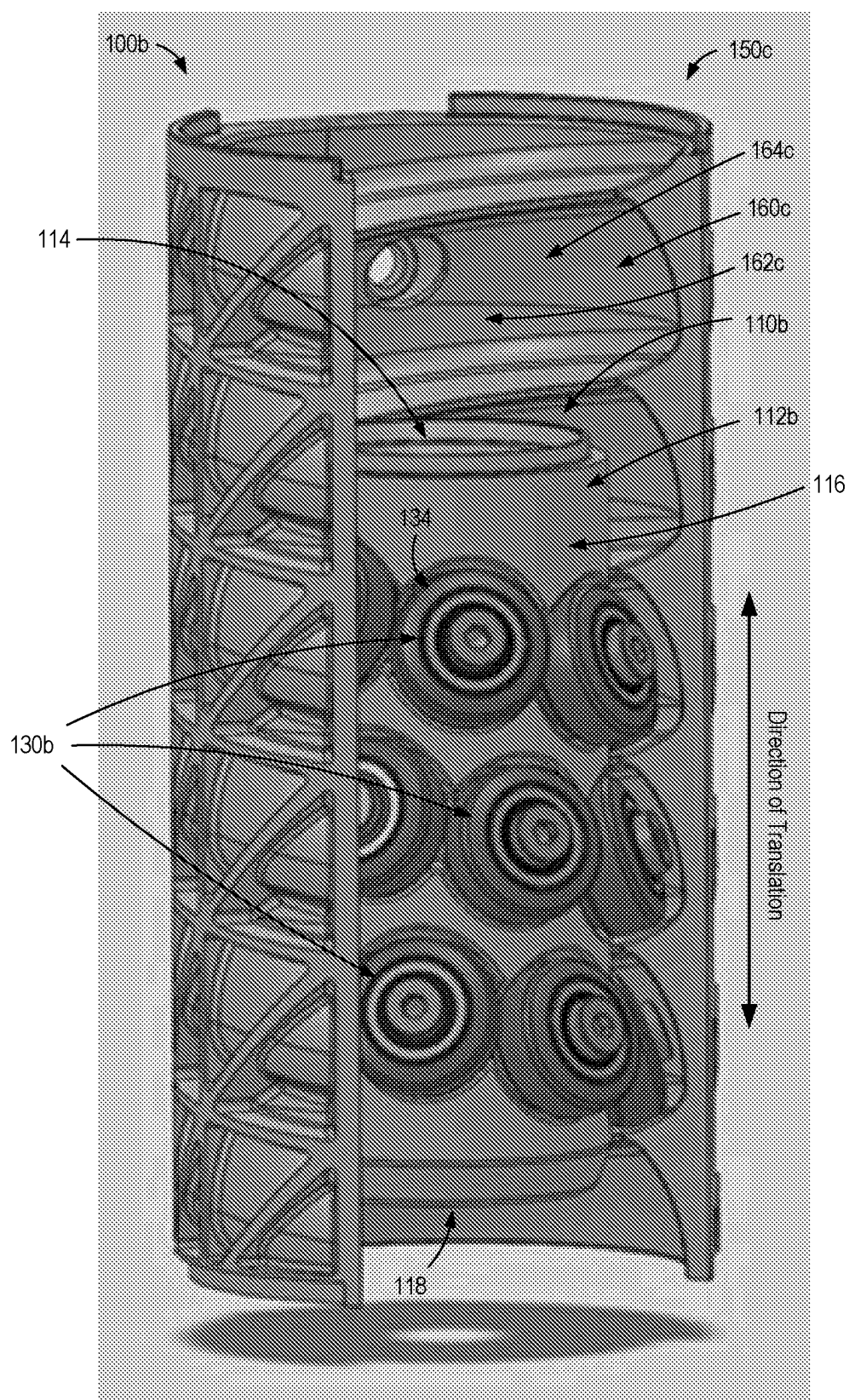


FIG. 3A

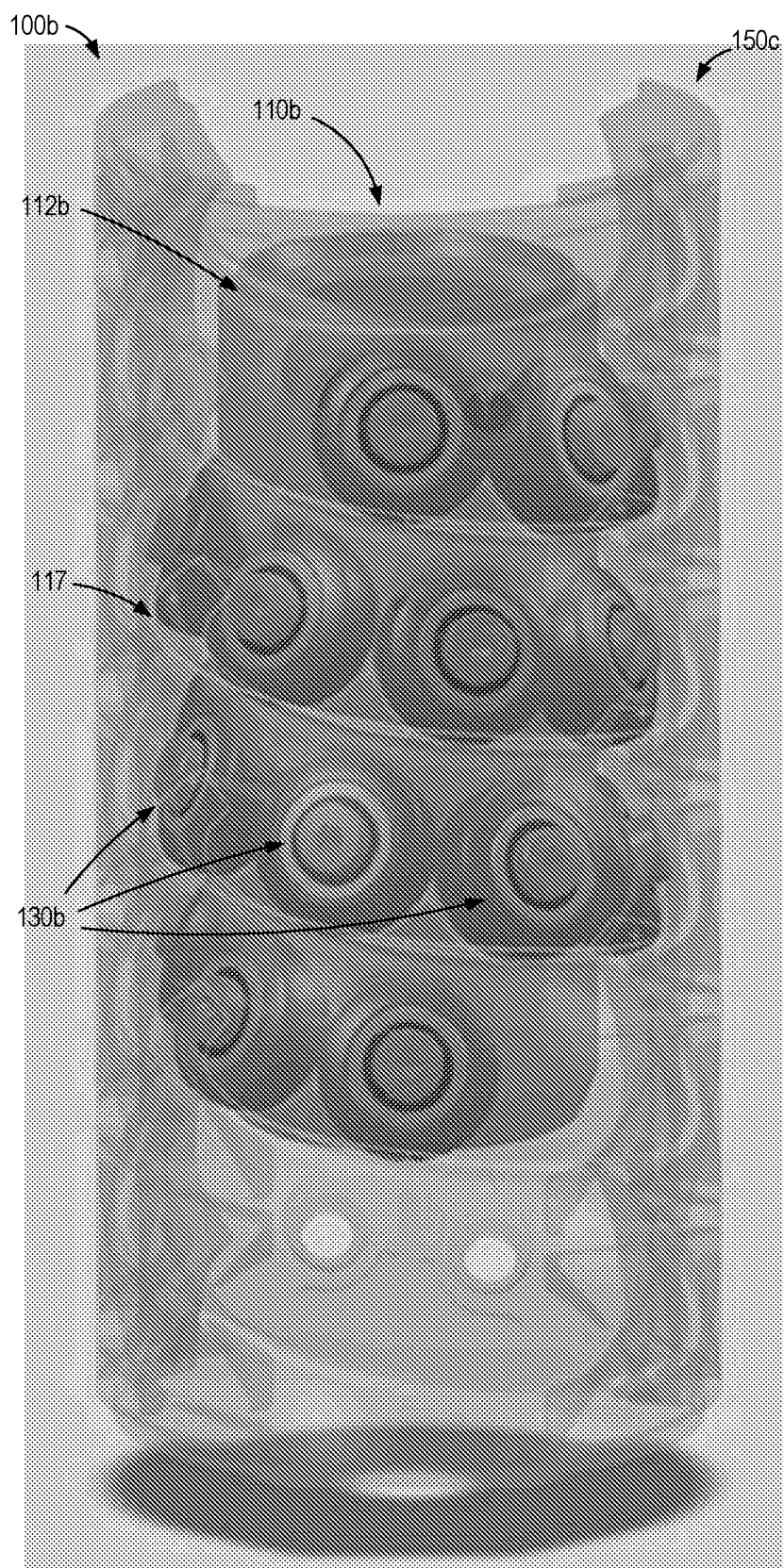


FIG. 3B

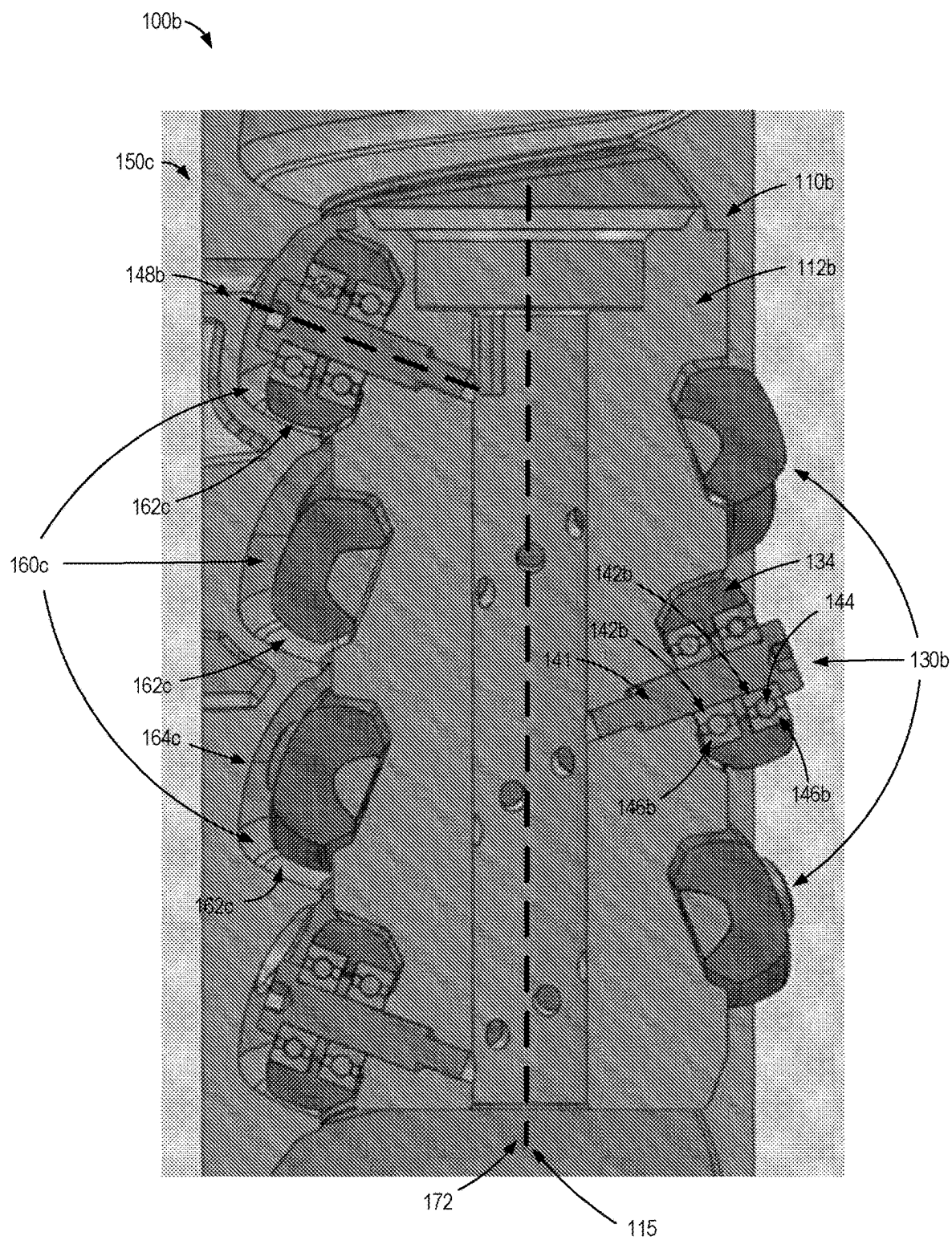


FIG. 3C

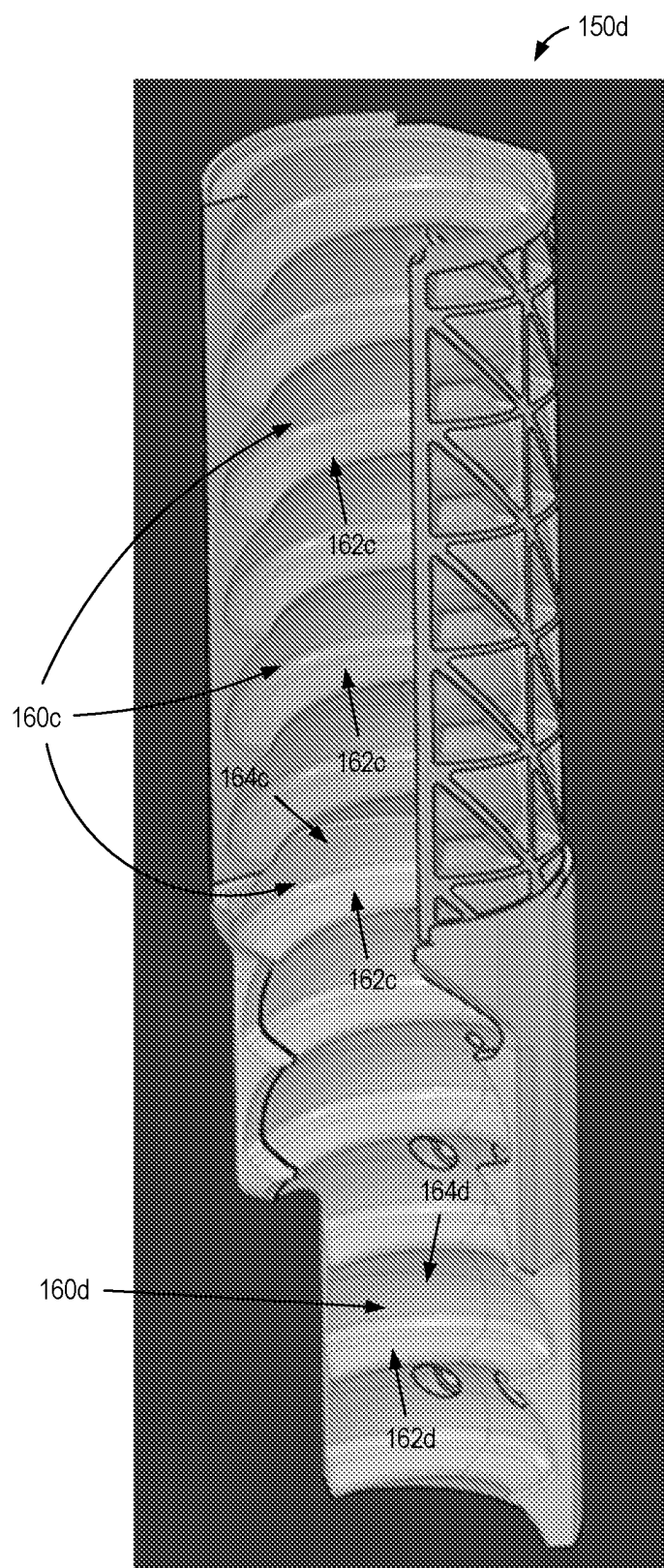


FIG. 4

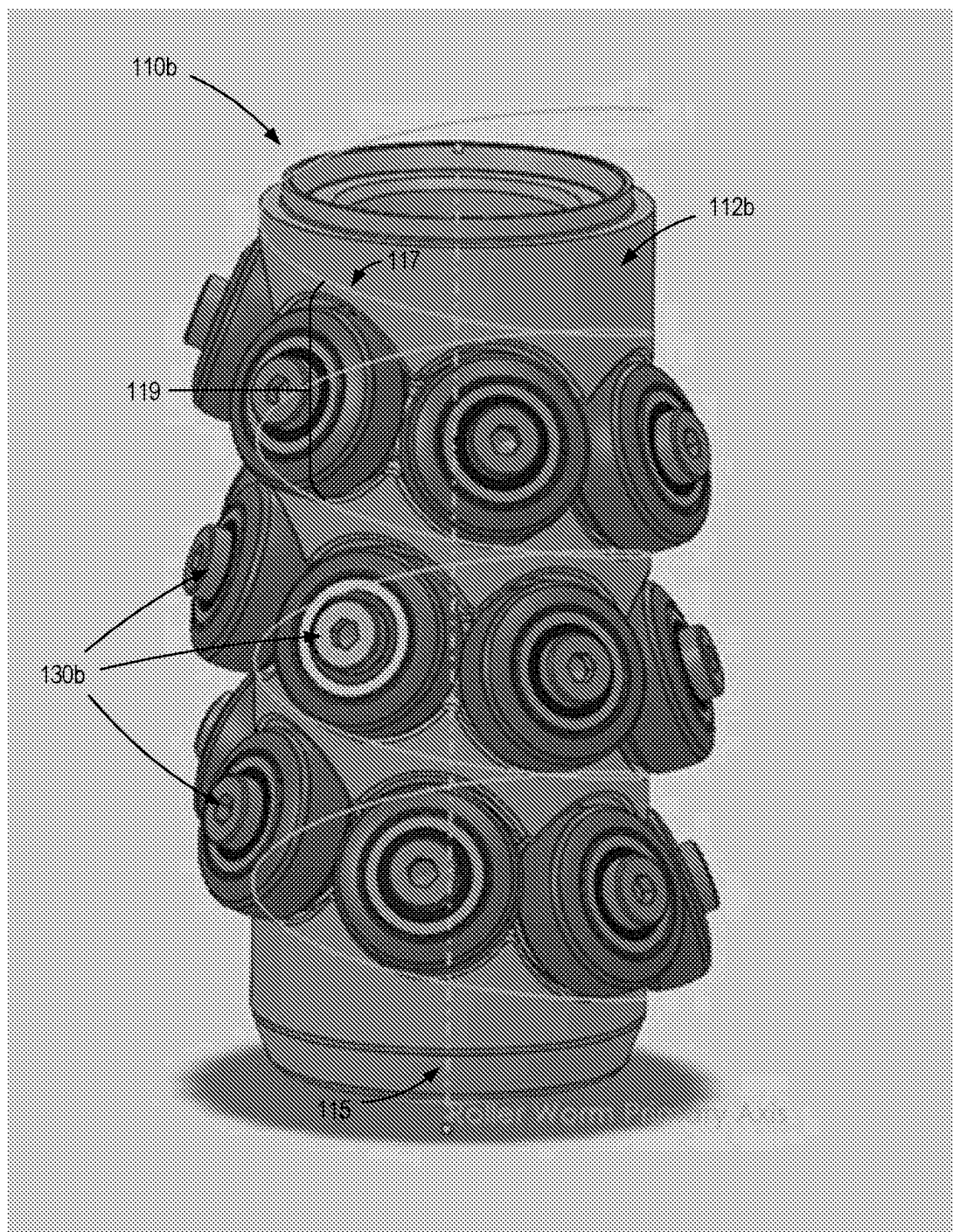


FIG. 5

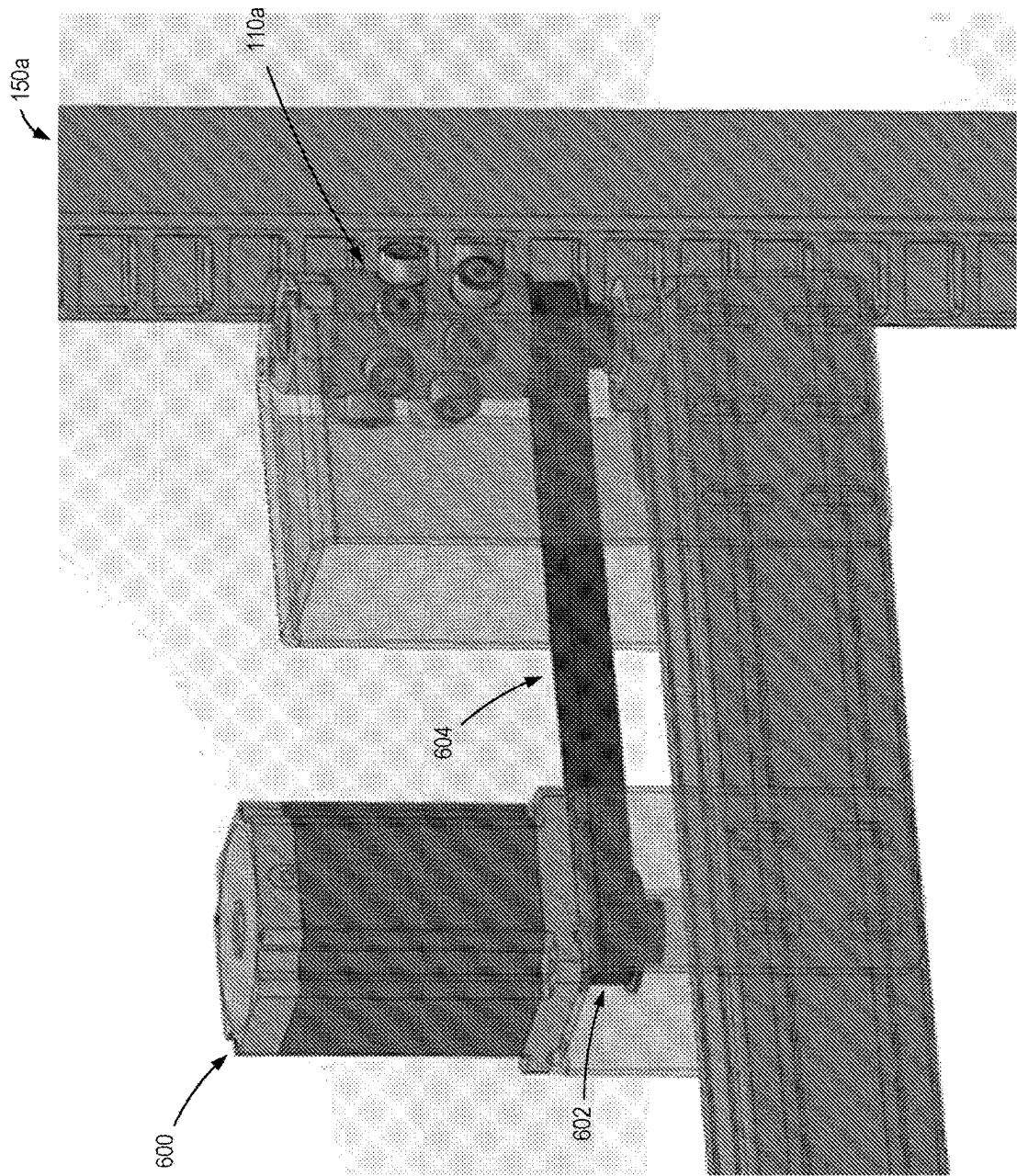


FIG. 6A

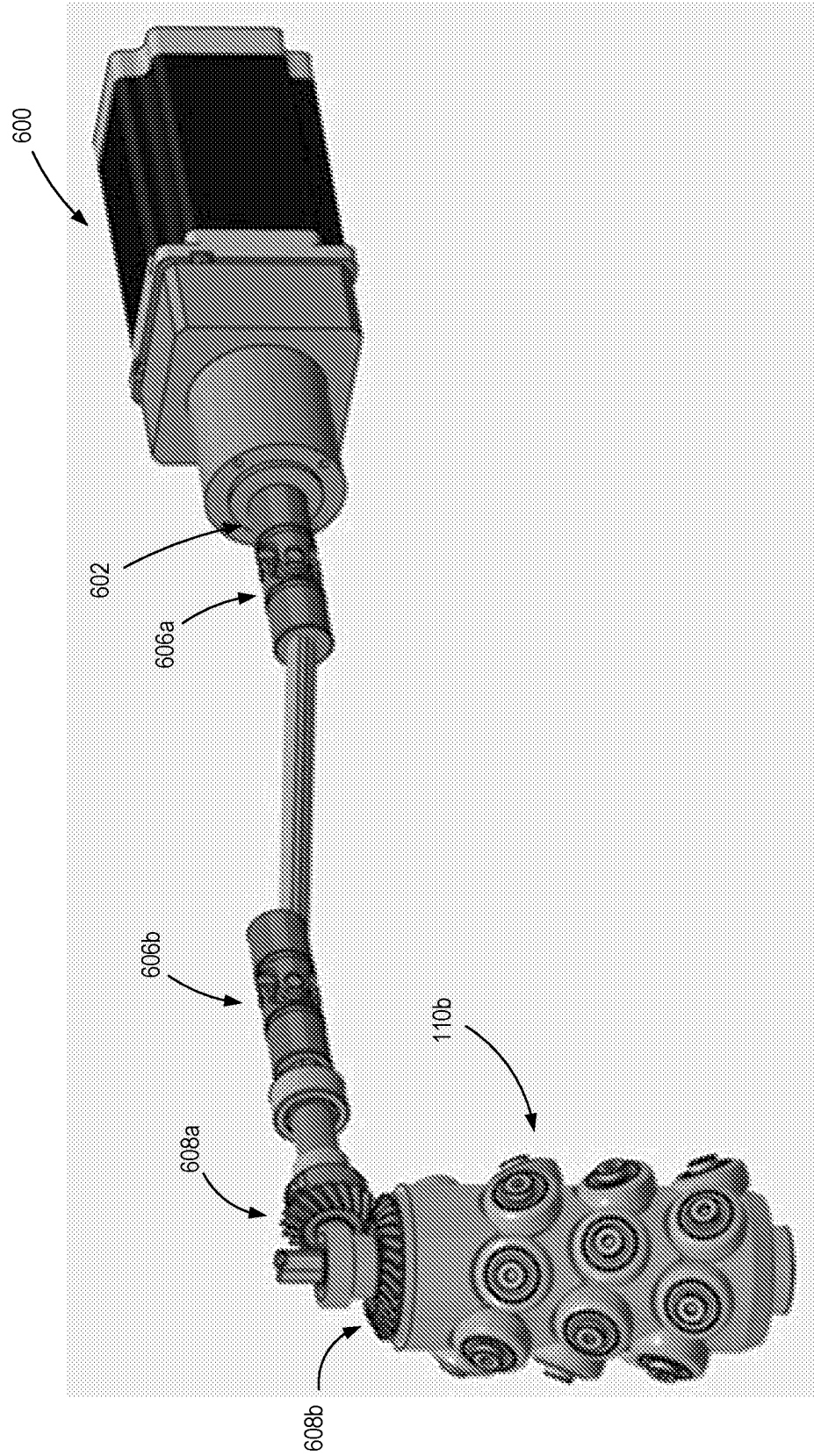


FIG. 6B

DISCRETE ROLLER ASSEMBLY-BASED HELICAL DRIVE FOR TRANSLATION

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 18/531,184, filed 6 Dec. 2023, which claims the benefit of and priority to U.S. Provisional Application Ser. No. 63/430,544, entitled “DISCRETE BEARING-BASED WORM GEAR FOR TRANSLATION,” filed on 6 Dec. 2022, each of which is incorporated herein in its entirety by this reference.

TECHNICAL FIELD

[0002] The following disclosure is directed to methods and systems for robotic movement, more specifically, methods and systems for operation and assembly of a discrete roller assembly-based helical drive used for translation along an axis.

BACKGROUND

[0003] A conventional threaded rod and nut mechanism can provide a motion transmission that can be used to mechanically translate a payload (e.g., a robot) along an axis. When the nut is split radially into a number of sections, each section could act upon the threaded rod (also referred to as a “threaded shaft” and “screw”) similar to a conventional worm gear by contacting and applying force to the nut at only a portion of the thread’s circumference. As an example, a worm gear mated to a rack to form a conventional worm drive can be used to move a robot coupled to the worm gear upward or downward along a vertical axis in a controlled manner. Such a conventional worm gear can produce a sliding frictional drag force (e.g., friction between a worm gear and a toothed wheel or frame) during operation of the worm gear. In a conventional worm drive, the sliding frictional force is proportional to a torque transmitted by the worm gear to the rack. Further, using a worm gear to lift a payload (e.g., a robot included in an automated storage and retrieval system (AS/RS)) relative to a frame (e.g., rack) can generate an additional fixed frictional drag force from a weight of the payload. This frictional drag force can contribute large power losses and provide added wear to worm drive components. By conservation of energy, this power is transformed primarily into heat, which can contribute to material property degradation and risk human safety associated with devices and/or machines that incorporate the worm drive components.

[0004] In some cases, replacing sliding friction elements of the worm drive with rolling elements can reduce frictional losses and wear of components involving sliding friction. Conventional solutions to the shortcomings involved with sliding frictional mechanisms can include replacing sliding friction elements with ball screws and/or roller screws, where individual rolling elements are contained between a nut and threaded rod. As an example, ball bearings included in a ball screw may contact and roll between the screw and nut components of the ball screw, where the ball bearings are recirculated in a ball return tube coupled to the nut during relative motion between the screw and the nut. However, such solutions are unable to accommodate engagement and disengagement between a screw and a section of a nut as

described with respect to a worm drive, which may be necessary based on an application of the nut, threaded rod, and rolling elements.

SUMMARY

[0005] To address the aforementioned shortcomings, a discrete roller assembly-based helical drive system used for translation along an axis is provided. Further, systems and methods incorporating the discrete roller assembly-based helical drive are provided. In one embodiment, a roller worm configured for translation along a primary axis is provided. In one aspect, the roller worm can include a body portion, and a plurality of discrete roller assemblies disposed in a first helical pattern about the body portion, where the roller worm is configured to rotate about a central axis to cause linear translation of the roller worm along a length of an engagement structure and the primary axis.

[0006] The roller worm may also include where the body portion includes a cylindrical structure including a first circular base, a second circular base, and a central body portion connecting the first circular base to the second circular base. The roller worm may also include where the central axis is parallel to and collinear with the primary axis. The roller worm may also include where the central axis is not parallel to the primary axis. The roller worm may also include where the roller assemblies have at least one of a cylindrical structure, a conical structure, or an ellipsoid structure. The roller worm may also include where the roller assemblies are disposed in the first helical pattern for at least one turn about the body portion. The roller worm may also include where the roller assemblies extend from a surface of the body portion along a respective roller assembly axis at an angle greater than 0 degrees and less than 180 degrees relative to the central axis. The roller worm may also include where a ratio of a pitch of the first helical pattern to a diameter of the body portion is in a range of 0.3 to 3.2. The roller worm may also include where a ratio of a diameter of each of the roller assemblies to a diameter of the body portion is in a range of 0.2 to 0.6. The roller worm may also include where at least one of the roller assemblies includes a roller element on an exterior of the roller assembly, where the roller element is configured to rotate about a roller assembly axis. The roller worm may also include where the engagement structure includes a plurality of tracks disposed along the length of the engagement structure. The roller worm may also include where a drive component is configured to cause rotation of the roller worm. The roller worm may also include where at least one supporting element is configured to cause the roller worm to linearly translate along the primary axis based on rotation of the roller worm about the central axis.

[0007] In some aspects, the roller worm may also include where the central axis extends through a center of each of the first circular base and the second circular base. The roller worm may also include where a surface of the central body portion is parallel to the central axis. The roller worm may also include where the at least one of the roller assemblies further includes (i) at least one inner race coupled to the body portion and (ii) at least one outer race rotationally coupled to the at least one inner race and configured to rotate about the roller assembly axis, where the roller element is coupled to the at least one outer race or the at least one outer race includes the roller element. The roller worm may also include where each of the tracks is configured to receive at

least one of the roller assemblies. The roller worm may also include where each of the tracks includes a complementary geometry relative to the roller assemblies. The roller worm may also include where each of the tracks includes at least one contact surface configured to contact at least one of the roller assemblies. The roller worm may also include where the roller assemblies are configured to (i) rotate about the central axis and (ii) contact the contact surfaces to cause translation of the roller worm along the length of the engagement structure. The roller worm may also include where the tracks are disposed in a second helical pattern along the length of the engagement structure, where a first pitch of the first helical pattern is equivalent to a second pitch of the second helical pattern. The roller worm may also include where the tracks are evenly spaced along the length of the engagement structure.

[0008] In another embodiment, a helical drive system is provided. In one aspect, the helical drive system can include a roller worm including a body portion, and a plurality of discrete roller assemblies disposed in a first helical pattern about the body portion. The helical drive system can also include an engagement structure including a primary axis, where the roller worm is configured to rotate about a central axis to cause linear translation of the roller worm along a length of the engagement structure and the primary axis.

[0009] In some aspects, the helical drive system may also include where the body portion includes a cylindrical structure including a first circular base, a second circular base, and a central body portion connecting the first circular base to the second circular base. The helical drive system may also include where the central axis is parallel to and collinear with the primary axis. The helical drive system may also include where the central axis is not parallel to the primary axis. The helical drive system may also include where the roller assemblies have at least one of a cylindrical structure, a conical structure, or an ellipsoid structure. The helical drive system may also include where the roller assemblies are disposed in the first helical pattern for at least one turn about the body portion. The helical drive system may also include where the roller assemblies extend from a surface of the body portion along a respective roller assembly axis at an angle greater than 0 degrees and less than 180 degrees relative to the central axis. The helical drive system may also include where a ratio of a pitch of the first helical pattern to a diameter of the body portion is in a range of 0.3 to 3.2. The helical drive system may also include where a ratio of a diameter of each of the roller assemblies to a diameter of the body portion is in a range of 0.2 to 0.6. The helical drive system may also include where at least one of the roller assemblies includes a roller element on an exterior of the roller assembly, where the roller element is configured to rotate about a roller assembly axis. The helical drive system may also include where the engagement structure includes a plurality of tracks disposed along the length of the engagement structure. The helical drive system may also include where a drive component is configured to cause rotation of the roller worm. The helical drive system may also include where at least one supporting element is configured to cause the roller worm to linearly translate along the primary axis based on rotation of the roller worm about the central axis.

[0010] In some aspects, the helical drive system may also include where the central axis extends through a center of each of the first circular base and the second circular base. The helical drive system may also include where a surface

of the central body portion is parallel to the central axis. The helical drive system may also include where the at least one of the roller assemblies further includes (i) at least one inner race coupled to the body portion and (ii) at least one outer race rotationally coupled to the at least one inner race and configured to rotate about the roller assembly axis, where the roller element is coupled to the at least one outer race or the at least one outer race includes the roller element. The helical drive system may also include where each of the tracks is configured to receive at least one of the roller assemblies. The helical drive system may also include where each of the tracks includes a complementary geometry relative to the roller assemblies. The helical drive system may also include where each of the tracks includes at least one contact surface configured to contact at least one of the roller assemblies. The helical drive system may also include where the roller assemblies are configured to (i) rotate about the central axis and (ii) contact the contact surfaces to cause translation of the roller worm along the length of the engagement structure. The helical drive system may also include where the tracks are disposed in a second helical pattern along the length of the engagement structure, where a first pitch of the first helical pattern is equivalent to a second pitch of the second helical pattern. The helical drive system may also include where the tracks are evenly spaced along the length of the engagement structure.

[0011] The above and other preferred features, including various novel details of implementation and combination of elements, will now be more particularly described with reference to the accompanying drawings and pointed out in the claims. It will be understood that the particular methods and apparatuses are shown by way of illustration only and not as limitations. As will be understood by those skilled in the art, the principles and features explained herein may be employed in various and numerous embodiments.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The disclosed embodiments have advantages and features which will be more readily apparent from the detailed description, the appended claims, and the accompanying figures (or drawings). A brief introduction of the figures is below.

[0013] FIG. 1A illustrates a first perspective view of an exemplary discrete roller assembly-based helical drive including an exemplary roller worm and an exemplary engagement structure, according to some embodiments.

[0014] FIG. 1B illustrates a second perspective view of an exemplary discrete roller assembly-based helical drive including an exemplary roller worm and an exemplary engagement structure, according to some embodiments.

[0015] FIG. 1C illustrates a third perspective view of an exemplary discrete roller assembly-based helical drive including an exemplary roller worm and an exemplary engagement structure, according to some embodiments.

[0016] FIG. 1D illustrates a fourth perspective view of an exemplary discrete roller assembly-based helical drive including an exemplary roller worm and an exemplary engagement structure, according to some embodiments.

[0017] FIG. 1E illustrates a cross-sectional view of an exemplary discrete roller assembly-based helical drive including an exemplary roller worm and an exemplary engagement structure, according to some embodiments.

[0018] FIG. 2 illustrates a perspective view of an exemplary engagement structure, according to some embodiments.

[0019] FIG. 3A illustrates a first perspective view of an exemplary discrete roller assembly-based helical drive including an exemplary roller worm and an exemplary engagement structure, according to some embodiments.

[0020] FIG. 3B illustrates a second perspective view of an exemplary discrete roller assembly-based helical drive including an exemplary roller worm and an exemplary engagement structure, according to some embodiments.

[0021] FIG. 3C illustrates a cross-sectional view of an exemplary discrete roller assembly-based helical drive including an exemplary roller worm and an exemplary engagement structure, according to some embodiments.

[0022] FIG. 4 illustrates a perspective view of an exemplary engagement structure, according to some embodiments.

[0023] FIG. 5 illustrates a perspective view of an exemplary discrete roller assembly-based roller worm, according to some embodiments.

[0024] FIG. 6A illustrates a perspective view of an exemplary drive component and an exemplary helical drive including an exemplary roller worm and an exemplary engagement structure, according to some embodiments.

[0025] FIG. 6B illustrates a perspective view of an exemplary drive component and an exemplary discrete roller assembly-based roller worm, according to some embodiments.

DETAILED DESCRIPTION

[0026] The figures (FIGS.) and the following description relate to some embodiments by way of illustration only. It is to be noted that from the following description, alternative embodiments of the structures and methods disclosed herein will be readily recognized as viable alternatives that may be employed without departing from the principles of the present disclosure.

[0027] Reference will now be made in detail to specific embodiments, examples of which are illustrated in the accompanying figures. It is to be noted that wherever practicable, similar or like reference numbers may be used in the figures and may indicate similar or like functionality. The figures depict some embodiments of the disclosed structures or systems (or methods) for purposes of illustration only. One skilled in the art will readily recognize from the following description that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles described herein.

[0028] As described herein, a conventional worm drive produces a sliding frictional drag force via friction between a worm gear and a toothed wheel or frame configured to interface with threads of the worm gear. Further, using a worm drive to lift a payload relative to a frame can generate an additional frictional drag force based on the payload's added weight. Solutions to reduce the frictional forces of conventional worm drives include use of rolling elements between a screw and threaded nut. However, as described above, these conventional mechanisms still have certain limitations, such as increased frictional drag forces (e.g., causing increased wear on included components) and an inability to accommodate engagement and disengagement between a screw and a section of a threaded nut.

[0029] The present disclosure addresses the aforementioned problems and other problems of existing mechanical translation systems by providing a discrete roller assembly-based roller worm configured for translation along an axis via contact with an engagement structure (e.g., a rack or frame), where the discrete roller assembly-based roller worm and engagement structure form a helical drive. The discrete roller assembly-based roller worm (referred herein to as a "roller worm") may include a cylindrical (e.g., substantially cylindrical) body portion (e.g., core) with discrete roller assemblies disposed about the body portion in a helical pattern that replace a continuous, helical thread of a conventional worm gear. The discrete roller assemblies may be (i) structured and arranged to provide a complementary surface between the body portion of the roller worm and an engagement structure (e.g., a frame or rack), (ii) structured and arranged to accommodate engagement and disengagement between the roller worm and the engagement structure, and (iii) constrained to a surface of the roller worm.

[0030] In some embodiments, an engagement structure may be a linear rack, where the roller worm is configured to translate along a length of the rack. The linear rack may include varying geometry along the length of the rack to enable engagement and disengagement between the discrete roller assemblies of the roller worm and the rack and/or to contain (e.g., capture) a roller worm and prevent disengagement between the roller worm and the rack. In some cases, the rack may include a number of concave structures configured to receive the roller assemblies of the roller worm, such that the concave structures are complementary to the structures of the discrete roller assemblies. The discrete roller assemblies of the roller worm may be (i) fixed to a surface of the body portion of the roller worm and (ii) configured to engage with and rotate along a surface of the rack to generate a linear thrust force, thereby causing linear translation of the roller worm along an axis corresponding to a length of the rack. The linear thrust force may act on a roller worm-to-rack interface and may be normal to a contact surface of the rack at which at least one discrete roller assembly contacts the rack. While the engagement structure is described herein as having a linear structure, the engagement structure may be a curved structure, thereby enabling arcuate translation of the roller worm along the engagement structure.

[0031] In some embodiments, motion of the roller worm may include and/or be constrained to (i) rotation about an axis (e.g., a primary, central axis) of the roller worm and translation along an axis (e.g., a primary axis) of the rack. By constraining motion of the roller worm as described herein, rotation of the roller worm about its central axis can cause the roller worm to linearly translate along the axis of the rack (e.g., corresponding to a length of the rack) via complementary forces. In some cases, a drive component (e.g., an external motor) may cause rotation of the roller worm about an axis (e.g., central axis) of the roller worm, thereby causing the roller worm to linearly translate along an axis of the rack.

[0032] In some embodiments, the axis about which the roller worm rotates and the axis along which the roller worm linearly translates may be oriented in parallel. In other cases, the axis about which the roller worm rotates may be oriented at an angle (e.g. ranging from greater than 0 degrees (0) to less than 90°) relative to the axis along which the roller

worm linearly translates, such that the axes are not oriented in parallel. The axes may be angled for manufacturing purposes and/or other considerations of a system (e.g., translation system) including the roller worm and the rack.

[0033] Referring to FIGS. 1A-1E and 3A-3C, views of exemplary helical drive systems **100a** and **100b** (collectively referred to herein as a helical drive system **100**) including a roller worm **110** and an engagement structure **150** are illustrated. In some embodiments, the roller worm **110** can include a body portion **112**. In some cases, the body portion **112** of the roller worm **110** may have a cylindrical (e.g., substantially cylindrical) structure including a first circular base portion **114**, a second circular base portion **118**, and a central body portion **116** connecting the first circular base portion **114** and the second circular base portion **118**. As an example, the body portion **112** may have right cylindrical shape or an oblique cylindrical shape. In some cases, when the body portion **112** has a cylindrical structure, a diameter of the body portion **112** may be in the range of approximately 50 mm to 70 mm. While the body portion **112** is shown and described as having a cylindrical structure (e.g., substantially cylindrical) formed from the first circular base portion **114**, the second circular base portion **118**, and the central body portion **116**, the body portion **112** may have an alternative structure such as an ellipsoid structure. In some cases, the roller worm **110** may be configured to rotate about a central axis **115**. In some cases, the central axis **115** may extend through a center of each of the first circular base portion **114** and the second circular base portion **118**. In some cases, a surface of the central body portion **116** may be oriented parallel (e.g., substantially parallel (to the central axis **115**)).

[0034] In some embodiments, the roller worm **110** may include a number of roller assemblies **130**. For example, the roller worm **110** may include N number of roller assemblies **130**, where N is any positive integer. In some cases, the roller assemblies **130** may have a cylindrical shape (e.g., substantially cylindrical shape). In other cases, the roller assemblies **130** may be conical-shaped roller assemblies or ellipsoid-shaped roller assemblies. Each of the roller assemblies **130** may be connected to the central body portion **116** of the roller worm **110**. The roller assemblies **130** may extend from a surface of the central body portion **116** in one or more helical turns around the central body portion **116**, such that the roller assemblies **130** form a helical (also referred to as “spiral”) pattern about the central body portion **116**. The roller assemblies **130** may extend from a surface of the central body portion **116** at an angle ranging from greater than 0° to less than 180° relative to the central axis **115**. As an example, referring to the helical drive system **100a** illustrated in FIGS. 1A-1E, a roller worm **110a** may include a body portion **112a** having roller assemblies **130a** each extending along a respective central axis **148a** orthogonally (90°) from a central body portion **116** of the body portion **112a** relative to the central axis **115**, where the roller assemblies **130a** each include a roller element **134**, an inner race **142a**, and an outer race **146a** rotationally coupled to the inner race **142a** and fixedly combined with the roller element **134** as described further herein. In some cases, the roller assemblies **130** may only extend toward the first circular base portion **114** or a second circular base portion **118** at an angle relative to the central axis **115**. As an example, referring to the helical drive system **100b** illustrated in FIGS. 3A-3C, a roller worm **110b** may include a

body portion **112b** having roller assemblies **130b** each extending along a respective central axis **148b** at angle that is not orthogonal (e.g., 70°) from a central body portion **116** of the body portion **112b** relative to the central axis **115**, where the roller assemblies **130b** (i) are angled toward the first circular base portion **114**, (ii) each include first and second inner races **142b**, (iii) each include first and second outer races **146b**, (iv) each include one or more roller bearing elements **144** disposed between the inner races **142b** and the outer races **146b**, where an outer race **146b** is rotationally coupled to a respective inner race **142** via one or more of the roller bearing elements **144**, and (v) each include a roller element **134** coupled to the exteriors of the first and second outer races **146b** as described further herein. The roller worm **110b** including roller assemblies **130b** each extending at an angle that is not orthogonal relative to the central axis **115** may enable (i) use of larger diameter roller assemblies **130b** relative to roller assemblies **130a** used in a roller worm **110a** and (ii) self-centering of the roller worm **110b** during translation along the axis **172**.

[0035] In some embodiments, the roller assemblies **130** may be disposed around the central body portion **116** in one or more helical turns having a selected (e.g., constant or variable) helical pitch. As an example, the roller assemblies **130** may be disposed around the central body portion **116** with a constant helical pitch of approximately 30 mm. A helical pitch of the one or more helical turns of the roller assemblies **130** may refer to a length between two adjacent positions on adjacent (e.g., consecutive) turns of the one or more helical turns along a central axis **115** of the roller worm **110**. For example, a helical pitch may be equivalent to a length between a center of a first turn of the roller assemblies **130** and a center of a second turn of the roller assemblies **130**, where the first turn and the second turn are adjacently positioned on the surface of the central body portion **116** and where an angular separation between the center of the first turn and the center of the second turn is 0° relative to a central axis **115** of the roller worm **110**. In some cases, the roller assemblies **130** may have a selected helical pitch without the roller worm **110** including more than one helical turn of roller assemblies **130**. Based on rotation of the roller worm **110** about the central axis **115**, the roller assemblies **130** may rotate about a helical path **117** and a helical pitch of the roller assemblies **130** may be equivalent to a distance **119** between adjacent points on consecutive helical turns of the roller assemblies **130** along the helical path **117** as illustrated in FIGS. 1B and 5. In some cases, the roller assemblies **130** may be evenly spaced or variably spaced along the helical path **117** on the body portion **112** of the roller worm **110**. As an example, the roller assemblies **130** may be evenly spaced along the central body portion **116**, such that each roller assembly **130** is a selected distance from another adjacent roller assembly **130**. As another example, the roller assemblies **130** may be variably spaced along the central body portion **116**, such that distances between adjacent roller assemblies **130** are not equivalent.

[0036] In some embodiments, a diameter of each of the roller assemblies **130**, a number of the roller assemblies **130** included in the roller worm **110**, and/or a helical pitch of the roller assemblies **130** may be selected based on a diameter of the body portion **112** of the roller worm **110**. As an example, for the roller worm **110**, a ratio of a helical pitch of the roller assemblies **130** and a diameter of the body portion **112** may range from approximately 0.3 to 3.2 and

may preferably be 0.6 (e.g., based on a ratio of a 30 mm helical pitch of the roller assemblies 130 and a 50 mm diameter of the body portion 112). As another example, a ratio of a diameter of each of the roller assemblies 130 and a diameter of the body portion 112 may range from approximately 0.2 to 0.6 and may preferably be 0.45 (e.g., based on a ratio of a 32 mm diameter of the roller assemblies 130 and a 70 mm diameter of the body portion 112).

[0037] In some embodiments, the roller assemblies 130 may be cam follower bearings (e.g., such as stud-mounted track rollers), deep groove ball bearings, or an alternative roller assembly. Each of the roller assemblies 130 may include a roller element 134 (e.g., having a cylindrical shape, crowned shape, conical shape, or ellipsoid shape) on an exterior of the roller assembly 130 to provide an interfacial surface between the roller worm 110 and an engagement structure 150. The roller element 134 may be fixedly coupled to the roller assembly 130 or rotationally coupled to the roller assembly 130 (e.g., such that the roller element 134 rotates about a central axis 148 of the roller assembly 130). Use of a roller element 134 that is configured to rotate about a central axis 148 can provide the advantage of reducing friction between the roller assembly 130 and the contact surfaces 162 of the engagement structure. In some cases, the roller element 134 may rotate about a central axis 148 of the roller assembly 130 and may contact the engagement structure 150 during translation of the roller worm 110 along the engagement structure 150. Each of the roller assemblies 130 may be made of and/or include one or more materials. For example, a roller assembly 130 may be made of and/or include a combination of engineering materials. In some cases, the central portion 141 may be made of and/or include steel and/or aluminum. In some cases, for a particular roller assembly 130, the inner race 142, the roller bearing element(s) 144, and/or the outer race 146 may be made of and/or include hardened steel, ceramic, plastic, and/or other engineering materials with suitable durability for engaging with the engagement structure 150 and enabling rotation of the outer race 146 and/or the roller element 134. In some cases, each of the roller assemblies 130 may have a diameter in the range of approximately 16 mm to 32 mm. In some cases, the roller assemblies 130 may be removably coupled to the body portion 112 by a central portion 141. For example, central portions 141 of the roller assemblies 130 may each be screwed into respective cavities (e.g., bores or holes) extending into the body portion 112 of the roller worm 110, such that the roller assemblies 130 are arranged in one or more helical turns around the body portion 112. In some cases, each of the roller assemblies 130 can include at least one inner race 142 coupled to the central portion 141, at least one outer race rotationally coupled to the at least one inner race 142, one or more roller bearing elements 144 that rotationally couple the at least one inner race 142 to the at least one outer race 146, and/or a roller element 134 coupled to an exterior of the outer race 146. In some cases, for a roller assembly 130, the outer race 146 may be combined with the roller element 134. For example, as illustrated in FIG. 1E, the outer race 146a is combined with the roller element 134, where both the roller element 134 and outer race 146a are configured to rotate about the central axis 148a.

[0038] In some cases, for each roller assembly 130 including an outer race 146, the inner race 142 may be rotationally coupled to the outer race 146, such that the outer race 146

and connected roller element 134 are configured to rotate about the central axis 148 via the inner race 142. In some cases, for each roller assembly 130, the outer race 146 and connected roller element 134 may rotate about a central axis 148 of the roller assembly 130 while the central portion 141 and inner race 142 remains fixed to the body portion 112 of the roller worm 110. For example, as illustrated in FIG. 3C, the roller element 134 is connected to exteriors of a pair of outer races 146b, where the outer races 146 are rotationally coupled to a pair of inner races 142b via one or more roller bearing elements 144, where both the roller element 134 and outer races 146b are configured to rotate about the central axis 148b. For each roller assembly 130, a central axis 148 of the roller assembly 130 about which the roller element 134 is configured to rotate may be collinear with an axis by which the roller assembly 130 extends from a surface of the central body portion 116 of the roller worm 110, where the roller assembly 130 extends from the surface of the central body portion 116 at an angle relative to the central axis 115 of the roller worm 110. Such rotation may reduce friction between the roller assembly 130 and a contact surface 162 of an engagement structure 150.

[0039] In some embodiments, the helical drive system 100 may include an engagement structure 150. In some cases, the engagement structure 150 may be a linear rack or a linear frame along which the roller worm 110 may linearly translate via rotation about the central axis 115 of the roller worm 110. When the roller worm 110 rotates about the central axis 115, contact (e.g., engagement) between the roller assemblies 130 and the engagement structure 150 may apply a linear thrust force to the roller worm 110, thereby causing the roller worm 110 to linearly translate along an axis 172 (also referred to as a “primary axis 172”) of the engagement structure 150. Based on a direction of rotation (e.g., clockwise or counterclockwise) of the roller worm 110 and an orientation of the helical pattern of the roller assemblies 130, the roller worm 110 may translate (e.g., linearly translate) in a first direction or second directions (e.g., forward or backward, left or right, or upward or downward) along a length of the engagement structure 150, where the first direction is opposite the second direction. In some cases, as illustrated in the FIGS. 1B, 1E and 3C, the central axis 115 about which the roller worm 110 rotates may be parallel to and collinear with the axis 172. In other cases, as illustrated in FIG. 1C, the central axis 115 about which the roller worm 110 rotates may not be parallel to the axis 172, such that the central axis 115 may be oriented at an angle relative to the axis 172 (e.g., ranging from greater than 0° to less than 90°).

[0040] In some embodiments, translation of the roller worm 110 may occur in a three-dimensional (3D) environment, where the axis 172 corresponds to any suitable axis along any axis (e.g., +/−, X, Y, Z). The engagement structure 150 may include a number of tracks 160 configured to receive and/or engage with the roller assemblies 130 of the roller worm 110. For example, the engagement structure 150 may include M number of tracks 160, where M is any positive integer. In some cases, each track 160 can include at least one complementary, contact surface 162 to the helical path 117 traveled by the roller assemblies 130 during rotation of the roller worm 110, where each contact surface 162 is configured to contact the roller elements 134 of each of the roller assemblies 130 of the roller worm 110 during translation of roller worm 110 along the engagement structure 150. In some cases, each track 160 may include a

concave structure (e.g., indentation) included in the engagement structure 150, where the concave structure includes at least one contact surface 162 and a distal end wall 164. A concave structure of a track 160 may be complementary with a geometry of the roller assemblies 130 and included roller elements 134 relative to the roller worm 110, such that the concave structure is configured to receive the roller assemblies 130. In some cases, the tracks 160 may be evenly spaced or variably spaced along the axis 172 and a length of the engagement structure 150. As an example, the tracks 160 and included contact surfaces 162 may be evenly spaced along the axis 172 and the length of the engagement structure 150, where a distance between centers of adjacent tracks 160 is based on (e.g., equivalent to) the helical pitch of the roller assemblies 130 of the roller worm 110. As another example, the tracks 160 and included contact surfaces 162 may be variably spaced along the axis 172 and the length of the engagement structure 150, where a distance between centers of adjacent tracks 160 is based on (e.g., equivalent to) the distances between adjacent roller assemblies 130 of the roller worm 110.

[0041] In some embodiments, during angular rotation of a roller assembly 130, the roller assembly 130 may engage with a track 160 (e.g., via contact with a contact surface 162 or entry into an indentation formed by the track 160) for a particular angular sweep (e.g., ranging from 0° to 360°) based on a geometry of the track 160. Accordingly, each track 160 may have a respective geometry configured to clear a particular angle swept by a roller assembly 130 as the roller assembly travels along the helical path 117 defined by rotation of the roller worm 110, where the angular sweep is defined relative to the axis 172. In some cases, each track 160 and included contact surface(s) 162 may have an angular sweep ranging from 300 to 3500 (e.g., preferably 270°), where the angular sweep is greater than an angular spacing between adjacent roller assemblies 130 of the roller worm 110. As an example, tracks 160a and 160b of the engagement structures 150a and 150b illustrated in FIGS. 1A-1E, respectively, may have an angular sweep of less than 180°. As another example, a track 160c of the engagement structure 150c illustrated in FIGS. 3A-3C may have angular sweep of greater than 180°.

[0042] In some embodiments, an engagement structure 150 may include tracks 160 having varying swept angles of contact surfaces 162 (e.g., to achieve different operating requirements for the helical drive system 100). For example, referring to the engagement structure 150d illustrated in FIG. 4, the tracks 160c including contact surfaces 162c may include an angular sweep greater than an angular sweep of the tracks 160d including the contact surface 162d. An engagement structure 150 including a track 160 and a contact surface 162 having a swept angle greater than 180° (e.g., half a full rotation of the roller worm 110) may be used to contain and guide the roller worm 110 along the axis 172. An engagement structure 150 including a track 160 and a contact surface 162 having a swept angle less than 180° may allow engagement and disengagement of the roller worm 110 from the engagement structure 150. In some cases, a particular track 160 may receive and/or contact one or more roller assemblies 130 (e.g. adjacent roller assemblies 130) at a point in time based on a size of the roller assemblies 130, a helical pitch of the roller assemblies 130, a distance (e.g., spacing) between adjacent roller assemblies 130, and/or an angular sweep of each of the tracks 160.

[0043] In some embodiments, as described herein and illustrated in FIGS. 1A-1E, the engagement structure 150 may include a number of tracks 160 configured to receive the roller assemblies 130. Each track 160 can include at least one contact surface 162 configured to engage with and contact (e.g., the roller elements 134 of the roller assemblies 130) the roller assemblies 130 of the roller worm 110. In some cases, a track 160 may include (i) only one contact surface 162 on a first side of a distal end wall 164 or (ii) a first contact surface 162 on a first side of a distal end wall 164 and a second contact surface 162 on a second side of the distal end wall 164, where the first side opposes the second side. As an example, as illustrated in FIGS. 1A and 1D-1E, the tracks 160a of the engagement structure 150a may each include a single contact surface 162a configured to engage with a first side of a roller element 134 of a roller assembly 130 of the roller worm 110. As another example, as illustrated in FIGS. 1B-1C, the tracks 160b of the engagement structure 150b may each include a pair of contact surfaces 162b on opposing interior sides of the track 160b separated by the distal end wall 164 and configured to engage with and contact first and second sides of a roller element 134 of a roller assembly 130 of the roller worm 110, where the contact surfaces 162b are positioned in parallel and where the first side of the a roller element 134 opposes the second side of the a roller element 134.

[0044] Referring to FIG. 2, an embodiment of the engagement structure 150a included in the helical drive system 100a is illustrated. For a use case of at least partially vertically translating (e.g., lifting) the roller worm 110 along the axis 172 and the length of the engagement structure 150a (e.g., against gravity), only a single contact surface 162a may be needed in each track 160a as illustrated in FIG. 2, as a lifting force acting on the roller worm 110 may only act in a direction resisting gravity (e.g., orthogonal to the contact surface 162a and/or parallel to the axis 172). In some cases, an engagement structure 150a may include only a single contact surface 162a per track 160a to (i) reduce cost and/or (ii) enable alternate methods for designing geometry with which the roller worm 110 and included roller assemblies 130 engage with the engagement structure 150a.

[0045] In some embodiments, each track 160 and included contact surface(s) 162 may be oriented at an angle ranging from greater than 0° to less than 180° relative to the axis 172 along which the roller worm 110 translates. As an example, referring to the engagement structures 150a and 150b as illustrated in FIGS. 1A-1E and 2, the tracks 160a and 160b, included contact surfaces 162a and 162b, and included distal end walls 164a and 164b, respectively, may be oriented orthogonally (90°) relative to the axis 172 along which the roller worm 110 translates. As another example, referring to the engagement structures 150c and 150d as illustrated in FIGS. 3A-3C and 4, respectively, each track (e.g., tracks 160c and 160d) and included contact surface(s) (e.g., contact surfaces 162c and 162d) and distal end walls (e.g., distal end walls 164c and 164d) may be helically oriented at an angle less than 90° or greater than 90° relative to the axis 172. Orienting the tracks 160 and included contact surfaces 162 at an angle less than 90° or greater than 90° relative to the axis 172 may provide an advantage of providing a centering geometry between the roller worm 110 and the engagement structure 150. In some cases, when the central axis 115 and the axis 172 are parallel, an offset distance can form between the central axis 115 and the axis 172 (e.g., such that the

central axis **115** and the axis **172** are not collinear). When the tracks **160** are oriented at an angle less than 90° or greater than 90° relative to the axis **172**, the non-perpendicular angle of each of the tracks **160** relative to the axis **172** can cause (i) contact between the track **160** and exteriors (e.g., the roller elements **134**) of the roller assemblies **130** in the direction of the offset distance and (ii) formation of a gap distance between the tracks **160** and exteriors of roller assemblies **130** oriented in a direction other than the direction of the offset distance. Based on the engagement structure **150** applying a force to the roller worm **110** that is normal to the contact surfaces **162** at the contact points between the roller worm **110** and engagement structure **150**, a first component of the force can act (e.g., upwards) on the roller worm **110** along the axis **172** and a second component of the force can act laterally on the roller worm **110** (e.g., in directions orthogonal to the axis **172**), thereby providing a centering force to return the central axis **115** to the axis **172** and reduce and/or eliminate the offset distance. A third component of the force may act orthogonally to the first and second components of the force based on the angle of the contact surface **162** with respect to the helical pattern of the roller assemblies **130** and the helical path **117**. The resultant of the first, second, and third components of the force can be a single vector directed normal to the contact surface **162**. When the tracks **160** and included contact surface(s) **162** are helically oriented at an angle less than 90° or greater than 90° relative to the axis **172**, a pitch of the tracks may be based on (e.g., equivalent to) a pitch of the roller assemblies **130** (e.g., as illustrated in FIGS. 3A-3C and 4).

[0046] In some embodiments, as described herein, the roller assemblies **130** may extend from a surface of the central body portion **116** at along a central axis **148** at an angle ranging from greater than 0° to less than 180° relative to the central axis **115** of the roller worm **110**. Accordingly, the engagement structure **150** may include tracks **160** and contact surfaces **162** that are complementary to a shape and/or an orientation of the roller assemblies **130**. The engagement structure **150** may be configured to (i) receive the roller assemblies **130** at an angle at which the roller assemblies **130** extend from the roller worm **110** relative to the central axis **115** and/or (ii) accommodate travel of the roller worm **110** and included roller assemblies **130** along the axis **172**. In some cases, the angle at which the tracks **160** receive the roller assemblies **130** may not be parallel to the angle at which the roller assemblies **130** extend from the roller worm **110** relative to the central axis **115** based on shapes of the exteriors (e.g., roller elements **134**) of the roller assemblies **130**. For example, when the roller assemblies **130** each include an roller element **134** having a conical shape, the tracks **160** may each have an angle relative to the central axis **115**, where the angle is a sum of the angle of the roller assemblies **130** and a taper angle of the conical shape of the roller element **134**. In some cases, each track **160** and included contact surface(s) **162** may oriented at an angle ranging from greater than 0° to less than 180° relative to the axis **172** along which the roller worm **110** translates, thereby enabling the track **160** to be oriented in parallel with the angle at which the roller assemblies **130** extend from the roller worm **110** relative to the central axis **115**. As an example, referring to the engagement structure **150a** as illustrated in FIGS. 1E, the tracks **160a** and included contact surfaces **162a** may be oriented (i) orthogonal (90°) relative to the axis **172** along which the roller worm **110a** translates

and (ii) parallel to the angle at which the roller assemblies **130a** extend from the roller worm **110a** relative to the central axis **115**. As another example, referring to the engagement structure **150c** as illustrated in FIG. 3C, the tracks **160c** and included contact surfaces **162c** may be oriented (i) at an angle greater than 0° and less than 90° or greater than 90° and less than 180° relative to the axis **172** which the roller worm **110b** translates and (ii) parallel to the angle at which the roller assemblies **130b** extend from the roller worm **110b** relative to the central axis **115**.

[0047] In some embodiments, an engagement structure **150** may be manufactured by stamping engagement structure **150** to form each track **160** and included contact surface(s) **162** and distal end wall **154**. For example, for an engagement structure **150a** including a single contact surface **162a** per track **160a** as illustrated in FIG. 2, the contact surface **162a** of each track **160a** may be fabricated by stamping a preform (e.g., made of and/or including sheet metal such as steel) to form the engagement structure **150a** and included contact surfaces **162a**. Such a fabrication technique for forming contact surfaces **162** on the engagement structure **150** may be optimized for cost and scalability. In some case, the tracks **160** and contact surfaces **162** along which the roller worm **110** translates may be formed by other manufacturing techniques such as cutting or injection molding.

[0048] In some embodiments, referring again to FIGS. 1A-1E and 3A-3C, each roller assembly **130** (e.g., the roller element **134** of each roller assembly **130**) may include crowning. Crowning included on the roller element **134** on the exterior of each roller assembly **130** may be (i) an interfacial surface between the roller assembly **130** and a contact surface **162** of the engagement structure **150** and (ii) configured to modify a location of a point of contact between each roller assembly **130** and the engagement structure **150**. When the roller assemblies **130** include crowning or on the roller elements **134**, during rotation of the roller worm **110** along the engagement structure **150**, a secondary radius of each of the crowned roller elements **134** (e.g., a crowning radius perpendicular to the radius of the roller assembly **130**) can provide a line of contact as the roller worm **110** rotates and sweeps each crowned roller element **134** along the helical path **117** across the surface of the engagement structure **150**. In some cases, additionally or alternatively, the roller element **134** may be a conical shaped object or ellipsoid shaped object to account for the helical path **117** of motion of the roller worm **110** across the surface of the engagement structure **150**. In some cases, the contact surfaces **162** of the tracks **160** may have a shape that is complementary to a shape of exteriors of the roller assemblies **130** (e.g., roller elements **134** of the roller assemblies **130**).

[0049] In some embodiments, each helical turn (e.g., revolution) of roller assemblies **130** around the central body portion **116** of the body portion **112** may include a number of roller assemblies **130** in a range of six to sixteen roller assemblies **130**. In some cases, each turn of the roller assemblies **130** around the central body portion **116** may preferably include at least eight roller assemblies **130**, such that each of at least eight roller assemblies **130** engage with a track **160** of the engagement structure **150** during a single revolution of the roller worm **110** along the helical path **117**. At least one of the roller assemblies **130** of the roller worm **110** may be in contact with a contact surface **162** the

engagement structure **150** during rotation of the roller worm **110** at any given time, such that at least part of the roller worm **110** is in continuous contact with the engagement structure **150**. In some cases, each turn of roller assemblies **130** around the central body portion **116** may include a non-integer number of roller assemblies **130**. For example, when the roller assemblies are evenly spaced, a first turn of roller assemblies **130** disposed about the central body portion **116** in a helical pattern may include 8.5 roller assemblies **130**, such that a second turn of roller assemblies **130** adjacent to the first turn of roller assemblies **130** shares half of a same roller assembly **130** with the first turn of roller assemblies **130**. Such a non-integer number of roller assemblies **130** per turn may be used to offset engagement and disengagement of individual roller assemblies **130** with the engagement structure **150**. A non-integer number of roller assemblies **130** per turn may be used to (i) reduce the swept angle of the tracks **160** of the engagement structure **150** and (ii) retain multiple points of contact between the roller worm **110** and the engagement structure **150** by reducing an angular separation (e.g., relative to a central axis **115**) between adjacent roller assemblies **130**.

[0050] Referring to FIGS. 6A and 6B, a perspective view of a drive component **600** and a roller worm **110** is illustrated. In some cases, as illustrated in FIGS. 6A and 6B, the roller worm **110** can be coupled to an output shaft **602** of a drive component **600**. The drive component **600** may be a motor or a motor and gearbox, where the output shaft **602** can cause rotation of the roller worm **110** about the central axis **115** of the roller worm **110**. In some cases, an external computing device and/or computing system communicatively connected to the drive component **600** may control the drive component **600** and subsequent rotation of the roller worm **110**. The computing device and/or computing systems may include at least one processor at least one memory storing computer-readable instructions that may be executed by the at least one processor (e.g., to control the drive component **600** and rotation of the roller worm **110**). In some cases, coupling of the output shaft **602** to the roller worm **110** may be accomplished by means of one or more gears, chains, belts, and/or shafts connecting the output shaft **602** to the roller worm **110a**. As an example, as illustrated in FIG. 6A, the roller worm **110a** can be coupled to the output shaft **602** by a belt **604** (e.g., a toothed timing belt). As another example, as illustrated in FIG. 6B, the roller worm **110b** can be coupled to the output shaft **602** by a pair of u-joints **606a** and **606b** and a pair of miter, bevel, and/or hypoid gears **608a** and **608b** configured to transmit torque between the drive component **600** and the roller worm **110b** (e.g., when the drive component **600** and the output shaft **602** are positioned orthogonally to the central axis **115** of the roller worm **110b**). In some cases, a drive component **600** can be positioned and mated with the roller worm **110** in a direct drive scenario, such that an axis of rotation of the output shaft **602** of the drive component **600** is collinear with and connected to the central axis **115** of the roller worm **110a**. In some cases, the drive component **600**, components connected to the drive component **600** (e.g., output shaft **602**, gears, chains, belts, etc.), and an optional gear reduction element can be housed within an interior cavity of the body portion **112** of the roller worm **110**, thereby enabling the roller worm **110** to rotate without use of a drive component **600** external to the roller worm **110**.

[0051] In some embodiments, coupling between the roller worm **110** and the output shaft **602** can be used to transmit torque bi-directionally between the drive component **600** and the roller worm **110**. As an example, to control a position of the roller worm **110a** along on the engagement structure **150a**, the drive component **600** and the output shaft **602** may transmit, via the belt **604**, torque to rotate the roller worm **110a**, thereby causing the roller worm **110a** to translate along the tracks **160a** of the engagement structure **150a** via the engagement between the roller assemblies **130** and the contact surfaces **162** of the tracks **160**. As another example, based on the forces (e.g., gravitational forces) acting on the roller worm **110a**, the roller worm **110a** may (i) rotate about the central axis **115**, (ii) translate (e.g., translate downward) along the engagement structure **150a**, and (ii) transmit a torque to the output shaft **602** and the drive component **600** based on the rotation and translation.

[0052] In some embodiments, any number of materials may be used for the body portion **112** of the roller worm **110** on which the roller assemblies **130** are coupled (e.g., mounted). In some cases, the body portion **112** may be made of and/or include steel, aluminum, brass, and/or engineering plastics. As an example, the body portion **112** may be made of steel for strength and durability purposes. In some cases, the engagement structure **150** may be made of and/or include steel, aluminum, brass, and/or engineering plastics. As an example, the engagement structure **150** may be made of steel for strength and processability (e.g., fabrication) purposes. In some cases, the roller elements **134** of the roller assemblies **130** may be made of and/or include steel, aluminum, brass, and/or engineering plastics. As an example, a roller element **134** may be comprised of steel for durability purposes and for having a closely matching coefficient of thermal expansion to the roller assembly **130**.

[0053] In some embodiments, one or more supporting and/or positioning elements may be used with the roller worm **110** to restrict relative movement between the roller worm **110** and the engagement structure **150**. The one or more supporting and/or positioning elements may restrict relative movement between the roller worm **110** and the engagement structure **150** to (i) rotational movement around the central axis **115** of the roller worm **110** and linear translation along the primary axis **172**, such that rotation of the roller worm **110** is converted to linear translation along the primary axis **172**. For example, one or more supporting and/or positioning elements may be configured to cause the roller worm to linearly translate along the primary axis **172** based on rotation of the roller worm about the central axis **115**. As illustrated in FIG. 6A, a supporting element may be fixed to both the roller worm **110** and the drive component to retain the roller worm **110** along the engagement structure **150**.

[0054] In some embodiments, a helical drive system **100** including a roller worm **110** and an engagement structure **150** may be applied for use in AS/RS systems. For example, a number of engagement structures **150** may be coupled to rails of a structure including a number of cuboidal cells. The cuboidal cells may be repeated to form a larger structure in a 3D array, so as to form a multi-level storage rack for item storage. To form the repeating structure, multiple cuboid cells can be aligned in rows, columns, and depth, which are also configurable, so that the repeating structure is designed to be modular for being easily adaptable to constraints in existing buildings, such as irregular walls, ceiling heights, or

building mechanical elements. A single material handling robot may be connected to and may control rotation of a number of roller worms **110** to navigate in a 3D environment from any cell to any cell, where the roller worms **110** are configured to engage with the engagement structures **150** that are connected to the rails forming the cells. According to some embodiments, an AS/RS system incorporating the roller worm **110** and engagement structure **150** may be a part of a material and conveyance and storage system that uses an automated material handling robot to store materials in a repeating structure (with repeated units such as cuboid cells) for retrieval at a later time. The automated material handling robot may transport the materials (also referred to as payload) by carrying the payload while traversing the repeating structure along any axis (e.g., +/-, X, Y, Z) from any cell to any adjacent cell. Here “+/-” may mean a forward/backward, left/right, or upward/downward navigation of a material handling robot along each X, Y, or Z axis in a 3D environment. Additional descriptions for an AS/RS system and related computing devices that can use and control the roller worm **110** and the engagement structure **150** is included in U.S. patent application Ser. No. 18/493,278, which is hereby incorporated by reference herein in its entirety.

Terminology

[0055] The phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

[0056] The term “approximately”, the phrase “approximately equal to”, and other similar phrases, as used in the specification and the claims (e.g., “X has a value of approximately Y” or “X is approximately equal to Y”), should be understood to mean that one value (X) is within a predetermined range of another value (Y). The predetermined range may be plus or minus 20%, 10%, 5%, 3%, 1%, 0.1%, or less than 0.1%, unless otherwise indicated.

[0057] The indefinite articles “a” and “an,” as used in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.” The phrase “and/or,” as used in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

[0058] As used in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the

contrary, such as “only one of or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

[0059] As used in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

[0060] The use of “including,” “comprising,” “having,” “containing,” “involving,” and variations thereof, is meant to encompass the items listed thereafter and additional items.

[0061] Use of ordinal terms such as “first,” “second,” “third,” etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed. Ordinal terms are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term), to distinguish the claim elements.

[0062] With respect to a roller worm as described herein, a “screw” gear often refers to translation of motion (e.g. lead screw) and a “worm” gear traditionally includes an involute-curve tooth profile. Based on a “roller screw” existing as a separate, pre-existing device, a roller worm may be the next most appropriate nomenclature as to not confuse with the existing roller screw.

We claim:

1. A system comprising:

a rack comprising a plurality of tracks, each sweeping a majority of an azimuthal range about a central axis; and a roller worm comprising:

a body defining a primary axis which is coaxial with the central axis; and

a plurality of discrete roller assemblies arranged helically about the primary axis of the body, at least one roller assembly contacting a track of the plurality.

2. The system of claim 1, wherein the roller worm is at least partially unenclosed.

3. The system of claim 1, wherein the roller worm is radially circumscribed by the rack along a majority of an azimuthal range of the roller worm about the primary axis and the roller worm is unenclosed by the rack along a remainder of the azimuthal range of the roller worm.

4. The system of claim 1, wherein each track of the plurality defines a sweep angle of less than 360 degrees about the central axis.

5. The system of claim 1, further comprising: a drive mechanism configured to rotate roller worm about the central axis to drive linear translation of the roller worm along the central axis.

6. The system of claim 5, wherein the drive mechanism is configured to apply force to the roller worm orthogonal to the primary axis.

7. The system of claim 6, wherein the at least one roller assembly defines a nonzero incline angle relative to the primary axis.

8. The system of claim 1, wherein the tracks are uniformly spaced along the central axis.

9. The system of claim 8, wherein the plurality of discrete roller assemblies defines a helical pattern with equal azimuthal spacing between adjacent roller assemblies.

10. The system of claim 1, wherein the discrete roller assemblies are arranged with equal azimuthal spacing between adjacent roller assemblies of the plurality of discrete roller assemblies.

11. The system of claim 1, wherein the plurality of roller assemblies has varying pitch relative to the primary axis.

12. The system of claim 11, wherein the at least one roller assembly comprises a convex engagement surface which contacts the track.

13. The system of claim 1, wherein the rack further comprises a second plurality of tracks, each sweeping a minority of the azimuthal range about the central axis.

14. A system comprising:

a roller worm body defining a primary axis and a reference plane orthogonal to the primary axis; and
a plurality of roller assemblies arranged helically about the primary axis of the roller worm body, each roller assembly independently coupled to the roller worm body and inclined relative to the reference plane.

15. The system of claim 14, wherein each roller assembly defines a respective rotational axis which is angled relative to the primary axis and has an orbital incline angle, relative to the reference plane, which is nonzero.

16. The system of claim 15, wherein the orbital incline angles of each roller assembly are equal.

17. The system of claim 14, wherein the plurality of roller assemblies defines a helical pattern with equal azimuthal spacing between adjacent roller assemblies of the helical pattern.

18. The system of claim 14, wherein the plurality of roller assemblies defines a helical pattern of varying pitch relative to the primary axis.

19. The system of claim 18, wherein each roller assembly comprises an ellipsoidal roller element.

20. The system of claim 18, wherein each roller defines a convex engagement surface.

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